<u>STRATEGIC SPECTRUM REFARMING FOR 5G DEPLOYMENT IN SRI LANKA</u>

MATH LAB ANALIZE

Code _ 01 : Coverage & Path Loss Comparison

```
% 5G Spectrum Refarming: Path Loss Comparison in Sri Lanka
clear; clc; close all;

% Frequencies (MHz)
f_700 = 700;    % 5G Sub-GHz (High Coverage)
f_2100 = 2100;  % 3G Legacy (Refarming target)
f_3500 = 3500;  % 5G C-Band (High Capacity)

% Distance array (from 0.1 km to 10 km)
d = 0.1:0.1:10;
c = 3e5; % Speed of light (km/s)

% Free Space Path Loss (FSPL) formula: FSPL = 20*log10(d) + 20*log10(f) + 32.44
FSPL_700 = 20*log10(d) + 20*log10(f_700) + 32.44;
FSPL_2100 = 20*log10(d) + 20*log10(f_2100) + 32.44;
FSPL_3500 = 20*log10(d) + 20*log10(f_3500) + 32.44;

% Plotting the results
figure('Name', 'Spectrum Propagation Analysis', 'Color', 'w');
plot(d, FSPL_700, 'LineWidth', 2.5, 'Color', '#0072BD'); hold on;
plot(d, FSPL_2100, 'LineWidth', 2.5, 'Color', '#EDB120');
plot(d, FSPL_3500, 'LineWidth', 2.5, 'Color', '#D95319');
```

```
grid on;

% Formatting the Graph
title('Free Space Path Loss Analysis: Refarming vs 5G Bands', 'FontSize', 14, 'FontWeight', 'bold');
xlabel('Distance from Base Station (km)', 'FontSize', 12, 'FontWeight', 'bold');
ylabel('Path Loss (dB)', 'FontSize', 12, 'FontWeight', 'bold');
legend('700 MHz (5G Coverage Layer)', '2100 MHz (Legacy 3G Phase-out)', '3.5 GHz (5G Capacity Layer)', 'Location', 'southeast', 'FontSize', 11);
set(gca, 'FontSize', 11, 'LineWidth', 1.2);
```

Code _ 02 : Network Capacity vs Bandwidth (Shannon Capacity Analysis)

```
% 5G Spectrum Refarming: Shannon Capacity Comparison
clear; clc; close all;

% Signal to Noise Ratio (dB)
SNR_dB = -10:2:30;
SNR_linear = 10.^(SNR_dB/10);

% Available Bandwidths for different generations (in Hz)
B_3G  = 15e6;   % 15 MHz (e.g., 2100 MHz band)
B_4G  = 20e6;   % 20 MHz (e.g., 1800 MHz band)
B_5G  = 100e6;  % 100 MHz (e.g., 3.5 GHz C-Band)
```

```
% Shannon Capacity Formula: C = B * log2(1 + SNR)
C_3G = (B_3G * log2(1 + SNR_linear)) / 1e6;  % Capacity in Mbps
C_4G = (B_4G * log2(1 + SNR_linear)) / 1e6;  % Capacity in Mbps
C_5G = (B_5G * log2(1 + SNR_linear)) / 1e6;  % Capacity in Mbps

% Plotting the results
figure('Name', 'Throughput & Capacity Analysis', 'Color', 'w');
plot(SNR_dB, C_3G, 'LineStyle', '--', 'LineWidth', 2.5, 'Color', '#77AC30'); hold on;
plot(SNR_dB, C_4G, 'LineStyle', '-.', 'LineWidth', 2.5, 'Color', '#0072BD');
plot(SNR_dB, C_5G, 'LineWidth', 3, 'Color', '#A2142F');
grid on;

% Formatting the Graph
title('Theoretical Network Capacity: 3G vs 4G vs 5G (Refarmed)', 'FontSize', 14, 'FontWeight', 'bold');
xlabel('Signal to Noise Ratio (SNR) in dB', 'FontSize', 12, 'FontWeight', 'bold');
ylabel('Channel Capacity (Mbps)', 'FontSize', 12, 'FontWeight', 'bold');
legend('3G Network (15 MHz Bandwidth)', '4G Network (20 MHz Bandwidth)', '5G C-Band (100 MHz Bandwidth)', 'Location', 'northwest', 'FontSize', 11);
set(gca, 'FontSize', 11, 'LineWidth', 1.2);
```

Code _ 03 : 5G Massive MIMO Beamforming

```
% 5G Infrastructure: Massive MIMO Beamforming Pattern (64 Elements)
clear; clc; close all;

N = 64;          % Number of antenna elements (Massive MIMO 64T64R)
d = 0.5;         % Distance between elements (half-wavelength)
theta = -90:0.1:90;  % Angle array (degrees)
theta_rad = theta * pi / 180;

% Target User angle (Beam Steering Angle)
theta_target = 30; % Pointing beam at 30 degrees
target_rad = theta_target * pi / 180;

% Array Factor Calculation
AF = zeros(1, length(theta));
for k = 1:N
    % Phase shift needed to steer the beam
    beta = -2 * pi * d * sin(target_rad);
    % Superposition of waves
    AF = AF + exp(1i * (k-1) * (2 * pi * d * sin(theta_rad) + beta));
end
```

```
% Normalize Array Factor
AF_normalized = abs(AF) / max(abs(AF));
AF_dB = 20*log10(AF_normalized);
AF_dB(AF_dB < -40) = -40; % Limit lower bound to -40dB for better visual

% Plotting Polar Pattern
figure('Name', '5G Massive MIMO Beamforming', 'Color', 'w');
polarplot(theta_rad, AF_dB + 40, 'LineWidth', 2, 'Color', '#7E2F8E');
ax = gca;
ax.ThetaZeroLocation = 'top';
ax.ThetaDir = 'clockwise';
ax.RTick = [0 10 20 30 40];
ax.RTickLabel = {'-40 dB', '-30 dB', '-20 dB', '-10 dB', '0 dB'};

% Formatting
title('5G Massive MIMO (64T64R) Beam Steering towards 30° User', 'FontSize', 13,
'FontWeight', 'bold');
```